Calculus 2 - Homework 4 (Problems)

1.
Use Simpson's Rule with n = 4 subintervals to approximate
integral_0^4 sqrt(x) dx. Round node values to 4 decimals and report S_4 to 4 decimals.

2.
Determine whether integral_0^1 1/sqrt(x) dx converges, and if so find its value.

3.
Use a comparison test to determine whether sum_{n=1}^infinity 1/(2^n + n) converges or diverges.

4.
Evaluate the definite integral integral_0^(pi/2) sin^2(x) cos^2(x) dx.

5.
Use a comparison test to determine whether sum_{n=1}^infinity 1/(n^2 + 1) converges or diverges.

6.
Evaluate the indefinite integral integral 2x(x^2 + 1)^5 dx.

7.
Evaluate integral 1/(x^2 + 4)^(3/2) dx.

8.
Determine whether integral_0^infinity e^(-2x) dx converges, and if so find its value.

9.
Classify sum_{n=1}^infinity (-1)^n/n^2 as absolutely convergent,
conditionally convergent, or divergent.

10.
Evaluate integral sin^3(x) cos^2(x) dx.